Practice Set A

Class 12 Mathematics, Chapter 4: Determinants

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, PYQ-style — 1 mark

If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then x is equal to:

(A) $\frac{12}{7}$ **(B)** 2 **(C)** -2 **(D)** $\frac{8}{7}$

2 MCQ, CBSE 2023 — 1 mark

Let A be a 3×3 matrix such that $|\text{adj } A| = 64$. Then $|A|$ is equal to:

(A) 8 only **(B)** -8 only **(C)** 64 **(D)** 8 or -8

3 MCQ, CBSE 2023 — 1 mark

If $|A| = 2$, where A is a 2×2 matrix, then $|4A^{-1}|$ equals:

(A) 4 **(B)** 2 **(C)** 8 **(D)** $\frac{1}{32}$

4 MCQ, CBSE SQP 2024-25 — 1 mark

If the points (x_1, y_1), (x_2, y_2) and $(x_1 + x_2,\ y_1 + y_2)$ are collinear, then $x_1 y_2$ is equal to:

(A) $x_2 y_1$ **(B)** $x_1 y_1$ **(C)** $x_2 y_2$ **(D)** $x_1 x_2$

5 MCQ, PYQ-style — 1 mark

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$, then the value of $a_{32} \cdot A_{32}$ is:

(A) 22 **(B)** -22 **(C)** 11 **(D)** -11

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** If A is a square matrix of order 3 and $\lvert A\rvert = 3$, then $\lvert\text{adj } A\rvert = 27$. **Reason (R):** For a square matrix A of order n, $\lvert\text{adj } A\rvert = \lvert A\rvert^{n-1}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the values of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$ and $(k, 4)$ is 35 sq units.	2 marks

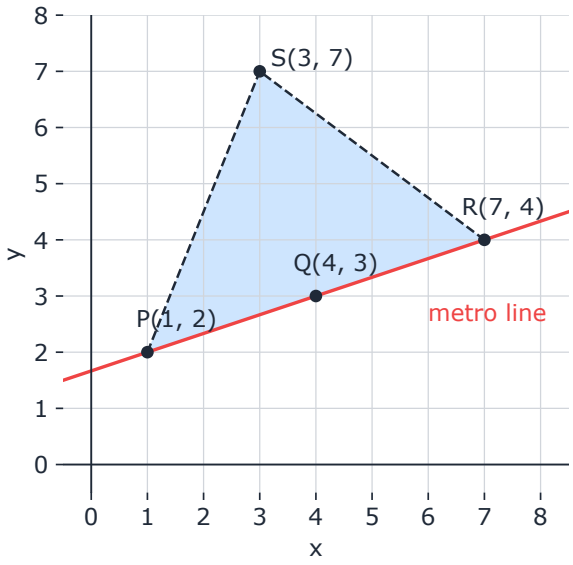
Level 2

17 marks

8	Short answer, PYQ-style Find the adjoint and the inverse of the matrix $A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$.	2 marks
9	Short answer, PYQ-style For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$. Hence find A^{-1}.	3 marks
10	Short answer, PYQ-style Using determinants, find the equation of the line joining $A(1, 3)$ and $B(0, 0)$. Hence find k if $D(k, 0)$ is a point such that the area of $\triangle ABD$ is 3 sq units.	2 marks
11	Short answer, PYQ-style Solve the following system of equations by the matrix method: $2x + 5y = 1$, $3x + 2y = 7$.	3 marks
12	Short answer, Board-pattern Examine the consistency of the system of equations: $2x - 3y = 5$, $6x - 9y = 15$.	2 marks
13	Short answer, Board-pattern If $A = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$, show that $A^{-1} = A$.	2 marks
14	Short answer, PYQ-style If $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.	3 marks

Q No.	Question	Marks
Level 3		**29 marks**
15	Long answer, CBSE SQP 2024-25 The equation of the path traversed by the ball headed by a footballer is $y = ax^2 + bx + c$ (where $0 \le x \le 14$ and $a, b, c \in \mathbb{R}$, $a \neq 0$) with respect to an XY-coordinate system in the vertical plane. The ball passes through the points $(2, 15)$, $(4, 25)$ and $(14, 15)$. Determine the values of a, b and c by solving the system of linear equations in a, b and c, using the matrix method. Also find the equation of the path traversed by the ball.	5 marks
16	Long answer, CBSE SQP 2025-26 For two matrices $A = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ -2 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & 0 & 3 \end{bmatrix}$, find the product AB and hence solve the system of equations: $3x - 6y - z = 3$, $2x - 5y - z + 2 = 0$, $-2x + 4y + z = 5$.	5 marks
17	Long answer, CBSE 2023 Solve the following system of equations by the matrix method: $x + 2y + 3z = 6$, $2x - y + z = 2$, $3x + 2y - 2z = 3$.	5 marks
18	Long answer, PYQ-style Gautam buys 5 pens, 3 bags and 1 instrument box and pays ₹160. From the same shop, Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays ₹190. Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays ₹250. Using the matrix method, find the cost of one pen, one bag and one instrument box.	5 marks
19	Long answer, CBSE 2025 Compartment If $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 2 \\ 7 & 3 & -3 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the system of equations: $3x + 4y + 7z = 14$, $2x - y + 3z = 4$, $x + 2y - 3z = 0$.	5 marks

Q No.	Question	Marks
20	Case study, Board-pattern **Case Study:** A city planner marks three metro stations on a map grid at $P(1,2)$, $Q(4,3)$ and $R(7,4)$. A new park is planned at $S(3,7)$ (1 unit $=100$ m).	4 marks



(i) Using determinants, show that the stations P, Q and R lie on one straight line. — 1 mark

(ii) Using determinants, find the area of the triangular plot with corners P, R and S. — 1 mark

(iii) Using determinants, find the equation of the line through P and R. — 2 marks

Answer key

Chapter 4: Determinants, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) 2
2	(D) 8 or -8
3	(C) 8
4	(A) x_2y_1
5	(A) 22
6	(D) Assertion (A) is false, but Reason (R) is true.
7	$k = 12$ or $k = -2$

Level 2

8	$\operatorname{adj} A = \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$, $A^{-1} = \frac{1}{14}\begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$
9	$a = -4, b = 1$; $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$
10	$y = 3x$; $k = \pm 2$
11	$x = 3, y = -1$
12	Consistent, with infinitely many solutions.
13	$A^{-1} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} = A$
14	$(AB)^{-1} = B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$. Verified.

Level 3

15	$a = -\frac{1}{2}, b = 8, c = 1$; path: $y = -\frac{1}{2}x^2 + 8x + 1$
16	$AB = I$; $x = 2, y = -3, z = 21$
17	$x = 1, y = 1, z = 1$
18	Pen ₹10, bag ₹20, instrument box ₹50
19	$A^{-1} = \frac{1}{62}\begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$; $x = 1, y = 1, z = 1$
20	(i) Determinant $= 0$, so collinear; (ii) 13 sq units; (iii) $x - 3y + 5 = 0$